

10-09-23 Mathematics

Ans. 1

$$\begin{aligned}16 + 8 - (7p + 3) \\ 24 - 7p - 3 \\ 21 - 7p = 0 \\ -7p = -21 \\ p = 3\end{aligned}$$

1) 3

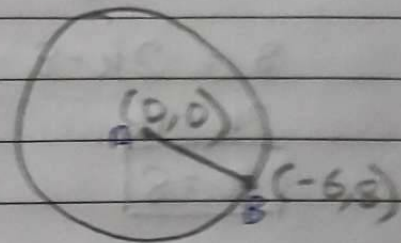
Ans. 2 4) no solutions

Ans. 3. radius = AB

$$\begin{aligned}&= \sqrt{(+6)^2 + (-8)^2} \\&= \sqrt{36 + 64} \\&= \sqrt{100}\end{aligned}$$

radius = 10 units

1) 10 units



Ans. 4 2) Proportional

Ans. 5. $\frac{\sqrt{3} \sin \theta}{\cos \theta} = 2 \sin \theta$

$$\frac{\sqrt{3} \tan \theta}{\sin \theta} = \frac{2}{\sqrt{3}}$$

$$\frac{1}{\cos \theta} = \frac{2}{\sqrt{3}}$$

$$\cos \theta = \frac{\sqrt{3}}{2} = \cos 30$$

$$\sin^2 30 - \cos^2 30$$

$$\frac{1}{4} - \frac{3}{4} = -\frac{2}{4} = -\frac{1}{2}$$

2) $-\frac{1}{2}$

Ans. 6. 1) $x^3 y^3$

Ans. 7 $\alpha + \beta = \frac{-b}{a} = \frac{2}{4} = \frac{1}{2}$

$\alpha\beta = \frac{c}{a} = \frac{k-4}{4}$

A/q.

$\frac{1}{2} = \frac{k-4}{4} \times \frac{1}{2}$

$\frac{1}{2} = \frac{k-4}{8}$

$8 = 2k - 8$

$2k = 16$

$k = 8$

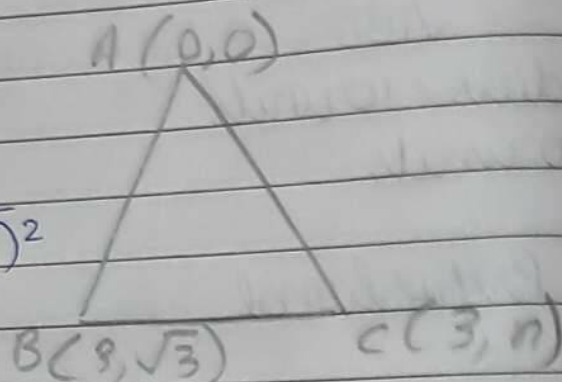
Ans. 8. as ABC is equilateral,

$AB = AC$,

$\sqrt{(-3)^2 + (-\sqrt{3})^2} = \sqrt{(k-3)^2 + (-x)^2}$

$9+3 = k^2 - 6k + 9 + x^2$

$x = 3$



Ans.

Ans. 9. For eq to have one solution,

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

In $2px + 3y = 7$,

$2x + y = 6$

$a_1 = 2p, a_2 = 2,$

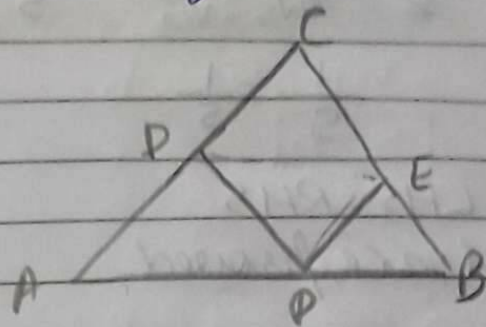
$b_1 = 3, b_2 = 1,$

$\Rightarrow \frac{2p}{2} \neq \frac{3}{1}$

$p \neq 3$

P shouldn't be 3 but can be any other number.

Ans. 10. Given,
 $PD \parallel BC$,
 $\frac{AD}{DC} = \frac{CE}{BE}$



To prove,
 $PE \parallel AC$

Construction,
~~draw DE~~

Proof,

As $PD \parallel BC$,

$$\frac{AD}{DC} = \frac{AP}{PB} \quad [\text{BPT}] \quad \text{--- (i)}$$

Also,

$$\frac{AD}{DC} = \frac{CE}{BE} \quad [\text{given}] \quad \text{--- (ii)}$$

From eq (i) and (ii),

$$\frac{AP}{PB} = \frac{CE}{BE}$$

By converse of BPT,

$PE \parallel AC$

Hence, proved.

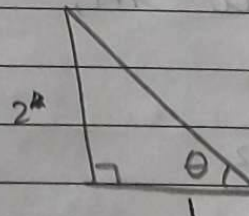
Ans. 11. $\tan \theta = \frac{1}{2} = \frac{P}{B}$

$$H = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\sin \theta = \frac{P}{H} = \frac{1}{\sqrt{5}}$$

$$\cos \theta = \frac{B}{H} = \frac{2}{\sqrt{5}}$$

$$\text{So, } \frac{1}{2} \sin \theta \cos \theta = \frac{1}{5}$$



$$\frac{1}{2} \left(\frac{1}{2\sqrt{5}} \right) \left(\frac{2}{\sqrt{5}} \right) = \frac{1}{5}$$

$$\frac{1}{5} = \frac{1}{5}$$

$$\text{LHS} = \text{RHS}$$

Hence, proved.

Ans. 12. $\frac{257}{5000} = \frac{257}{2^3 \times 5^4}$

It is a terminating decimal.

Ans. 13. To prove,
 $\sqrt{3}$ is irrational,
 Proof.

To the contrary, let's assume

$\sqrt{3}$ is rational i.e.,

$\sqrt{3} = \frac{a}{b}$ where a, b are co prime

$$3 = \frac{a^2}{b^2} \quad (\text{squaring both sides})$$

$$b^2 = \frac{a^2}{3}$$

a^2 is divisible by 3

so a is also divisible by 3,

Now, let $a = 3m$,

$$3 = \frac{9m^2}{b^2}$$

$$b^2 = \frac{9m^2}{3}$$

$$m^2 = \frac{b^2}{3}$$

b^2 is divisible by 3,

b is also divisible by 3

This contradicts the fact that a & b are co-prime.
This contradiction occurred due to our wrong assumption.

So,

$\sqrt{3}$ is irrational,
Hence, proved.

Ans. 14. $p(x) = 2x^2 - 3x + 7$

$$\alpha + \beta = \frac{-b}{a} = -\frac{(-3)}{2} = \frac{3}{2}$$

$$\alpha\beta = \frac{c}{a} = \frac{7}{2}$$

$$\frac{1}{2\alpha-3} + \frac{1}{2\beta-3}$$

$$\frac{(2\beta-3) + (2\alpha-3)}{(2\alpha-3)(2\beta-3)} = \frac{2\beta-3+2\alpha-3}{4\alpha\beta-6\alpha-6\beta+9}$$

$$= \frac{2\alpha+2\beta-6}{4\alpha\beta-6\alpha-6\beta+9}$$

$$= \frac{2(\alpha+\beta)-6}{14-6\left(\frac{3}{2}\right)+9} = \frac{2\left(\frac{3}{2}\right)-3}{14-9+9} = \frac{-3}{2} \quad \boxed{\frac{-3}{2}}$$

Ans. 15. For it to have infinite solutions,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$3x - (a+1)y = 2b-1$$

$$5x + (1-2a)y = 3b$$

$$a_1 = 3, b_1 = -(a+1), c_1 = -2b+1$$

$$a_2 = 5, b_2 = (1-2a), c_2 = -3b$$

$$\frac{3}{5} = \frac{-(a+1)}{1-2a}$$

$$3-6a = -5a-5$$

$$\boxed{8=a}$$

$$\frac{3}{5} = \frac{1-2b}{-3b}$$

$$-9b = 5 - 10b$$

$$\boxed{5 = b}$$

Ans. 16. Given,

$$\triangle ABE \cong \triangle ACD$$

To prove,

$$\triangle ADE \sim \triangle ABC$$

Proof.

$$\text{As } \triangle ABE \cong \triangle ACD$$

$$AB = AC \quad] \text{ [CPCT] } \text{--- (i)}$$

$$AD = AE \quad] \text{--- (ii)}$$

In $\triangle ADE$ and $\triangle ABC$,

dividing both eqs

$$\frac{AD}{AB} = \frac{AE}{AC}$$

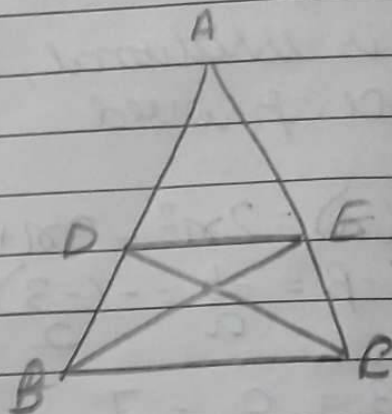
$$\frac{AD}{AB} = \frac{AE}{AC}$$

$$\angle A = \angle A \text{ (common)}$$

By SAS,

$$\boxed{\triangle ADE \sim \triangle ABC}$$

Hence, proved.



Ans. 17.

$$\frac{\tan^2 \theta}{\tan^2 \theta - 1} + \frac{\operatorname{cosec}^2 \theta}{\sec^2 \theta - \operatorname{cosec}^2 \theta}$$

$$= \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$$

$$\frac{\sin^2 \theta - 1}{\cos^2 \theta} + \frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}$$

$$= \frac{\sin^2 \theta \times \cos^2 \theta}{\cos^2 \theta \sin^2 \theta - \cos^2 \theta} + \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

$$\frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\sin^2 \theta \cos^2 \theta}{\sin^2 \theta (\sin^2 \theta - \cos^2 \theta)}$$

$$\frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta} = \text{RHS}$$

Hence, proved.

Ans. 18. Given,

$$C(1,1), A(-2,7)$$

$$m_1 = 3, m_2 = 2,$$

Using section formula,

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$1 = \frac{3(x_2) + (2)(-2)}{5}$$

$$5 = 3x_2 - 4$$

$$3x_2 = 9$$

$$\boxed{x_2 = 3}$$

~~mmmm~~

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

$$1 = \frac{(3)(y_2) + (2)(7)}{5}$$

$$5 = 3y_2 + 14$$

$$3y_2 = -9$$

$$\boxed{y_2 = -3}$$

$$\text{Coordinates of B} = \boxed{(3, -3)}$$

Ans. 19. Let fraction be $\frac{x}{y}$
Given,

$$\frac{2x}{y+1} = 1 \Rightarrow 2x = y+1 \quad \text{--- (1)}$$

$$\begin{array}{ccc} (-2, 7) & & (1, 1) \\ A & \xrightarrow{3} & C \xrightarrow{2} B \end{array}$$

$$\frac{x+y}{2y} = \frac{1}{2} \Rightarrow x+y=y \quad \text{--- (1)}$$

solving both eq.

$$2x - y - 1 = 0$$

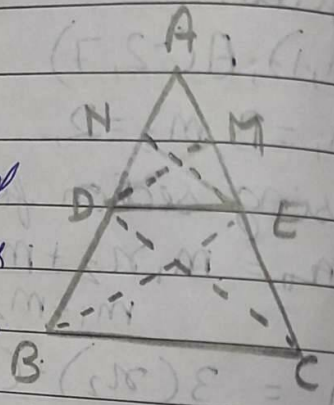
$$-x + y - 4 = 0$$

$$\boxed{x=5}$$

$$\boxed{y=9}$$

$$\text{fraction} = \frac{5}{9}$$

Ans. 20. Thales theorem - If a line is drawn parallel to one side of a triangle and intersects other two sides at distinct points, then other two sides are divided in same ratio.



Given,

ABC is a triangle

$DE \parallel BC$,

To prove,

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Construction

Join BE and CD,

draw $DM \perp AC$ and $EN \perp AB$,

Proof,

$$\text{ar}(\triangle ADE) = \frac{1}{2} \times AD \times EN$$

$$\text{ar}(\triangle BDE) = \frac{1}{2} \times BD \times EN$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{AD}{BD} \quad \text{--- (1)}$$

$$\frac{ar(\triangle ADE)}{ar(\triangle CDE)} = \frac{\cancel{DE} \times AE \times \cancel{DM}}{\cancel{DE} \times EC \times \cancel{DM}}$$

$$\frac{ar(\triangle ADE)}{ar(\triangle CDE)} = \frac{AE}{EC} \quad \text{--- (i)}$$

$$ar(\triangle BDE) = ar(\triangle CDE) \quad \text{--- (ii)}$$

→ [Δ 's on same base, between same parallel lines]

from eq (i), (ii) and (iii),

$$\boxed{\frac{AD}{DB} = \frac{AE}{EC}}$$

Hence, proved.

Ans. 21

$$\begin{aligned} & \frac{\sin 30^\circ + 2\cos^2(45^\circ) + \tan^2(60^\circ)}{\frac{1}{2}\cot 45^\circ + \cos^2(30^\circ) + \tan^2(45^\circ)} \\ &= \frac{1 + 2 \cdot \left(\frac{\sqrt{2}}{2}\right)^2 + \tan^2(60^\circ)}{\frac{1}{2}\cot 45^\circ + \cos^2(30^\circ) + \tan^2(45^\circ)} \end{aligned}$$

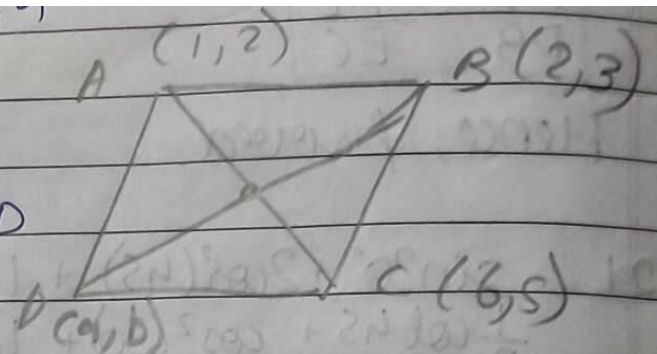
$$\begin{aligned} &= \frac{\frac{1}{2} + 2 \cdot 1 + (\sqrt{3})^2}{\frac{1}{2}\cot 45^\circ + \cos^2(30^\circ) + \tan^2(45^\circ)} \end{aligned}$$

$$\begin{aligned} &= \frac{9/2}{\frac{1}{2} + \frac{3}{4} + \tan^2(45^\circ)} \end{aligned}$$

$$= \boxed{2}$$

$$-10x - 6y = -4$$

Ans. 22. Join AC and BD,
O is midpoint of AC & BD
so,



midpoint of AC

$$\rightarrow \left(\frac{6+1}{2}, \frac{5+2}{2} \right) = \left(\frac{7}{2}, \frac{7}{2} \right)$$

midpoint of BD

$$\rightarrow \left(\frac{2+a}{2}, \frac{3+b}{2} \right)$$

As both points are same,

$$\frac{2+a}{2} = \frac{7}{2}$$

$$\boxed{a=5}$$

$$\frac{3+b}{2} = \frac{7}{2}$$

$$\boxed{b=4}$$

coordinates of D = (5, 4)

Ans. 23. A will complete 1 round = $\frac{1980}{330} = 6 \text{ min}$

B " " " = $\frac{1980}{198} = 10 \text{ min}$

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C will complete 1 round = $\frac{1980}{220} = 9 \text{ min}$
 let them meet after time t ,
 they've completed n_1, n_2 & n_3 rounds,
 all are x dist away from start point
 so,

$$330t = x + 1980n_1$$

$$198t = x + 1980n_2$$

$$220t = x + 1980n_3$$

$$\rightarrow 198t = 330t - 1980n_1 + 1980n_2$$

$$1980(n_1 - n_2) = 132t$$

$$t = 15(n_1 - n_2)$$

$$220t = 330t - 1980n_1 + 1980n_3$$

$$1980(n_1 - n_3) = 110t$$

As t is same and $n_1 - n_3$ or $n_1 - n_2$ can be whole no,

LCM of 15, 18

$$= \boxed{90 \text{ min}}$$

$\therefore$ They will meet after 90 minutes.

Ans. 24. let,

son's age = x

father's age = y

A/q,

$$2x + y = 70 \quad \text{--- (I)}$$

$$2y + x = 95 \quad \text{--- (II)}$$

Multiplying eq. (I) by 2 and subtracting from eq. (II),

$$2y + x = 95$$

$$-2y - 4x = -140$$

$$-3x = -45$$

$$\boxed{x = 15}$$

Putting in eq. (II),

$$2y = 80$$

$$\boxed{y = 40}$$

Ans. 25 Using section formula, $A \xrightarrow{k} O \xrightarrow{1} B$
 $(1, -7) \quad (n, 0) \quad (6, 4)$

$$y = \frac{m_1 y_2 + m_2 y_1}{k+1}$$

$$0 = \frac{(k)(4) + (1)(-7)}{k+1}$$

$$4k - 7 = 0$$

$$k = \frac{7}{4}$$

$$\text{ratio} = \frac{k}{1}$$

$$= \frac{7 \times 1}{4} = \boxed{7:4}$$

Ans. 26. $m^2 n = (\operatorname{cosec} \theta - \sin \theta)^2 (\sec \theta - \cos \theta)$
 $= \left(\frac{1 - \sin^2 \theta}{\sin \theta} \right)^2 \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right)$

Similarly,

$$mn^2 = (\operatorname{cosec} \theta - \sin \theta) (\sec \theta - \cos \theta)^2$$

$$= \frac{\sin^4 \theta \cos^2 \theta}{\cos^2 \theta \sin \theta} = \sin^3 \theta$$

$$(m^2 h)^{2/3} + (mn^2)^{2/3} = \sin^2 \theta + \cos^2 \theta$$

$$= \boxed{1}$$

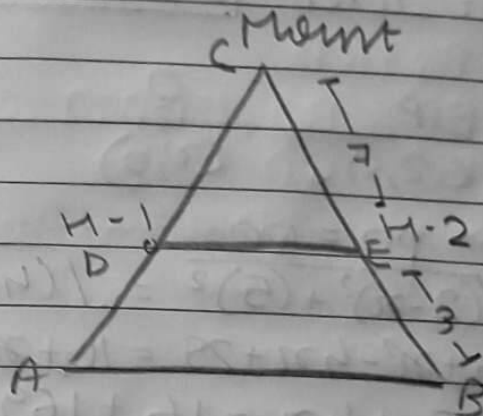
Hence, proved.

Ans. 27 i) clearly $DE \parallel AB$,
so, In $\triangle ACB$ and DCE ,
 $\angle C = \angle C$ (common)

$$\frac{AD}{DC} = \frac{BE}{EC}$$

By SAS, $\triangle ACB \sim \triangle DCE$

$$\frac{BC}{EC} = \frac{10}{7} = \boxed{10:7}$$



~~all sides~~

~~side~~

Ans. 28. i) Using section formula

$$P(2,5) \quad Q(x,y) \quad R(8,3)$$

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$x = \frac{(1)(8) + (2)(2)}{3} = \frac{8+4}{3} = \frac{12}{3}$$

$$\boxed{x=4}$$

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

$$y = \frac{(1)(3) + (2)(5)}{3} = \frac{3+10}{3} = \boxed{\frac{13}{3}}$$

coordinates of Q = $\left(4, \frac{13}{3}\right)$

ii) PR

$$\begin{aligned}
 &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(8 - 2)^2 + (3 - 5)^2} \\
 &= \sqrt{36 + 4} \\
 &= \sqrt{40} = \boxed{2\sqrt{10} \text{ units}}
 \end{aligned}$$

iii)

P (2, 5) O (x, 0) Q (4, $\frac{13}{3}$)

$$PO = QO$$

$$\sqrt{(2-x)^2 + (5)^2} = \sqrt{(4-x)^2 + \left(\frac{13}{3}\right)^2}$$

$$4 + x^2 - 4x + 25 = 16 + x^2 - 8x + \frac{169}{9}$$

$$29 + 4x = 16 + \frac{169}{9}$$

$$13 + 4x = \frac{169}{9}$$

$$52 + 16x = 169$$

$$16x = 117$$

$$x = \frac{117}{16}$$

$$4x = \frac{169 - 52}{9}$$

$$4x = \frac{52}{9}$$

$$x = \frac{52}{9} \cdot \frac{13}{13}$$

$$\boxed{x = \frac{13}{3}}$$

iv)

Midpoint of PR

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{10}{2}, \frac{8}{2} \right) = \boxed{(5, 4)}$$

Ans 29 Let speed of A = x km/hr
" " B = y km/hr

when travelling in same direction,
relative speed $\rightarrow x - y$

$$\text{dist} = 100 \text{ km}$$

$$t = 5 \text{ hr}$$

$$\text{dist} = S \times t$$

$$100 = (x - y)5$$

$$x - y = 20 \text{ — (i)}$$

when moving in opp. direction

relative speed $\rightarrow x + y$

$$\text{dist} = 100 \text{ km}$$

$$t = 1 \text{ hr}$$

$$\text{dist} = S \times t$$

$$100 = (x + y)$$

$$x + y = 100 \text{ — (ii)}$$

Solving both eq's

$$x - y = 20$$

$$x + y = 100$$

$$2x = 120$$

$$\boxed{\begin{array}{l} x = 60 \text{ km/hr} \\ y = 40 \text{ km/hr} \end{array}}$$